

1. APL ship

WHAT YOU SEE

A wrecked ship labeled APL listing and sinking into a blood-red sea — the foundering ship is acute promyelocytic leukemia, a maturation arrest at the promyelocyte stage.

WHAT IT ENCODES

Acute promyelocytic leukemia (FAB M3) is defined by t(15;17)(q24;q21) producing the PML::RARA fusion, which blocks myeloid maturation at the promyelocyte stage. Blasts/abnormal promyelocytes show heavy azurophilic granules and bundles of Auer rods (faggot cells), are strongly MPO-positive, and are characteristically CD34-/HLA-DR-negative on flow.

BOARD TRAP

APL is the one AML subtype that is a true medical emergency from the moment it is SUSPECTED — and it is the most curable; missing it early (death from hemorrhage) is the classic fatal error.

2. DIC bleeding sea

WHAT YOU SEE

Sheets of red blood pouring from the APL ship and staining the whole sea — the uncontrolled bleed-out is DIC plus hyperfibrinolysis consuming clotting factors.

WHAT IT ENCODES

APL classically presents with disseminated intravascular coagulation (DIC) plus hyperfibrinolysis: the abnormal promyelocytes release tissue factor and annexin II, consuming clotting factors and fibrinogen. Labs: low fibrinogen, low platelets, high D-dimer, prolonged PT/aPTT. This bleeding diathesis is the leading cause of early death.

BOARD TRAP

The APL emergency is COAGULOPATHY/hemorrhage (DIC + hyperfibrinolysis), not leukostasis — and the WBC is often low/normal, so don't be reassured by a low count.

3. Bleeding sites (GI/brain/lung)

WHAT YOU SEE

Spotlit casualties tagged GI, BRAIN, and LUNG bleeding around the ship — the three bleed sites are the feared DIC hemorrhages (intracranial being most lethal).

WHAT IT ENCODES

DIC-driven hemorrhage causes intracranial hemorrhage (the most feared, leading early-death cause), pulmonary hemorrhage, and GI bleeding. Early death from bleeding can occur before or during the first days of treatment, which is why blood-product support and ATRA must start immediately.

BOARD TRAP

A new headache or neuro change in a suspected APL patient is intracranial hemorrhage until proven otherwise — act on it emergently.

4. FFP / Cryo / Plt support

WHAT YOU SEE

Stacked transfusion bags labeled FFP, CRYO, PLT, and PRBC on the dock — the wall of blood products is the aggressive coagulopathy support APL demands.

WHAT IT ENCODES

Aggressive coagulopathy support is mandatory: transfuse to keep fibrinogen >100-150 mg/dL (cryoprecipitate/FFP) and platelets >30,000-50,000, especially while bleeding or pre-procedure. Continue until DIC resolves with differentiation therapy.

BOARD TRAP

Do not wait for active bleeding to transfuse in APL — prophylactically correct fibrinogen and platelets because spontaneous fatal hemorrhage is common.

5. ATRA man

WHAT YOU SEE

A figure glowing brilliant yellow pointing the way (ATRA) — the radiant 'vitamin A' man forces the arrested promyelocytes to mature and resolves the DIC.

WHAT IT ENCODES

All-trans retinoic acid (ATRA, a vitamin A derivative) binds the RARA portion of the fusion and forces terminal differentiation of the malignant promyelocytes, rapidly correcting the coagulopathy. Start ATRA on clinical SUSPICION of APL, before cytogenetic/molecular confirmation, because it both treats the leukemia and resolves the life-threatening DIC.

BOARD TRAP

Do NOT wait for cytogenetics/PCR to confirm t(15;17) before starting ATRA — empiric ATRA on suspicion saves lives and is reversible if APL is excluded.

6. Arsenic trioxide (green)

WHAT YOU SEE

A figure glowing toxic green beside ATRA (arsenic) — the green 'poison' man is arsenic trioxide, which degrades the PML::RARA fusion protein.

WHAT IT ENCODES

Arsenic trioxide (ATO) degrades the PML::RARA fusion protein. The combination ATRA + ATO is curative and chemo-FREE for low/intermediate-risk APL (Lo-Coco trial: superior to ATRA + chemo). ATO prolongs the QT interval — monitor ECG and replete K⁺/Mg²⁺.

BOARD TRAP

ATRA + ATO replaces cytotoxic chemo entirely in standard/low-risk APL; remember ATO's QT prolongation and electrolyte requirements.

7. ~85% cure

WHAT YOU SEE

An 85% sign posted on the safe dock away from the sinking ship — the high number marks APL as the most curable AML when treated promptly.

WHAT IT ENCODES

With prompt ATRA-based therapy, APL has the BEST prognosis of any AML subtype — cure rates ~85-90% (and >95% for low-risk treated with ATRA+ATO). The historically worst-prognosis leukemia became the most curable thanks to differentiation therapy.

BOARD TRAP

APL went from worst to best prognosis among AMLs — but only if treated promptly; early hemorrhagic death remains the main threat to that 85-90% cure rate.

8. WBC ≤10,000 low risk

WHAT YOU SEE

A gauge reading LOW RISK ≤10,000 WBC — the low-pinned dial is the Sanz threshold that selects the chemo-free ATRA+ATO arm.

WHAT IT ENCODES

Sanz risk stratification uses presenting WBC: WBC ≤10,000/μL = low/intermediate risk → treat with ATRA + ATO without cytotoxic chemotherapy.

BOARD TRAP

Risk in APL is defined by WBC count (and platelets), NOT by genetics — t(15;17) is present in all APL; the WBC threshold of 10,000 separates the treatment arms.

9. WBC >10,000 high risk

WHAT YOU SEE

A gauge reading HIGH RISK >10,000 WBC — the high-pinned dial is the Sanz threshold that adds cytotoxic chemo for cytoreduction.

WHAT IT ENCODES

WBC >10,000/ μ L at presentation = high-risk APL → add cytotoxic chemotherapy (an anthracycline such as idarubicin/daunorubicin, sometimes gemtuzumab ozogamicin) to ATRA ± ATO for cytoreduction and leukostasis prevention.

BOARD TRAP

High WBC APL needs added chemo/cytoreduction; this is also the group at highest risk for differentiation syndrome.

10. No chemo (low risk)

WHAT YOU SEE

A NO CHEMO placard on the gate beside the low-risk gauge — the crossed-out chemo sign means low-risk APL is cured with ATRA+ATO alone.

WHAT IT ENCODES

Standard/low-risk APL is cured with ATRA + arsenic trioxide and avoids cytotoxic chemotherapy entirely, sparing patients its toxicity and secondary-malignancy risk.

BOARD TRAP

Reflexively giving 7+3 to an APL patient is wrong — low-risk APL is a chemo-free cure with ATRA+ATO.

11. Differentiation syndrome bay

WHAT YOU SEE

A hospital bay signed DIFFERENTIATION SYNDROME with a febrile patient under a chest film showing infiltrates — the ward depicts the cytokine-surge complication of ATRA/ATO.

WHAT IT ENCODES

Differentiation (APL/retinoic acid) syndrome occurs within days-weeks of starting ATRA or ATO: fever, dyspnea, pulmonary infiltrates, pleural/pericardial effusions, peripheral edema, weight gain, hypotension, and acute kidney injury, driven by a cytokine surge from differentiating promyelocytes. Higher risk with rising WBC.

BOARD TRAP

New dyspnea, edema, or pulmonary infiltrates on ATRA/ATO = differentiation syndrome, NOT pneumonia or CHF — and steroids, not antibiotics/diuretics alone, are the answer.

12. Dexamethasone

WHAT YOU SEE

A medication label reading DEX 10 Q12H in the differentiation-syndrome bay — the steroid order is the first-line treatment for differentiation syndrome.

WHAT IT ENCODES

Treat differentiation syndrome with dexamethasone 10 mg IV q12h promptly at first suspicion; in severe cases temporarily hold ATRA/ATO. Prophylactic steroids or prophylactic cytoreduction may be used in high-WBC patients.

BOARD TRAP

Do not stop differentiation therapy permanently for mild differentiation syndrome — add dexamethasone and continue/resume ATRA once controlled.

13. PCR monitoring

WHAT YOU SEE

A benchtop PCR machine beside the clinician — the molecular analyzer is RT-PCR tracking PML::RARA transcripts for residual disease and relapse.

WHAT IT ENCODES

Monitor PML::RARA transcripts by RT-PCR for measurable residual disease and molecular relapse. Molecular CR (PCR-negative) is the goal after consolidation; a rising/positive PCR signals impending relapse and prompts pre-emptive therapy.

BOARD TRAP

In APL, molecular (PCR) monitoring of PML-RARA — not just morphologic CR — drives relapse detection and consolidation decisions.

14. 10% mortality sign

WHAT YOU SEE

A 10% MORTALITY placard nailed to the differentiation-syndrome bay wall — the number is the early-death rate (hemorrhage/differentiation syndrome) in the first weeks.

WHAT IT ENCODES

Despite excellent overall cure rates, early death (mostly from hemorrhage, and from differentiation syndrome) still accounts for roughly 10% of APL mortality, concentrated in the first weeks — underscoring why immediate ATRA and aggressive coagulopathy support matter.

BOARD TRAP

The main cause of APL death is EARLY (hemorrhage/differentiation syndrome in the induction window), not late relapse — prognosis hinges on surviving the first weeks.

15. APL syndrome 3-week timer

WHAT YOU SEE

An APL SYNDROME 3 WEEKS timer near the high-risk gauge — the countdown marks the ~1-3 week window when differentiation syndrome typically strikes.

WHAT IT ENCODES

Differentiation syndrome characteristically arises within the first ~1-3 weeks of starting ATRA/ATO; vigilance is highest during this early window when the WBC is rising.

BOARD TRAP

Differentiation syndrome is a time-limited early complication — falsely attributing week-2 dyspnea to infection delays the steroids that are actually needed.
